

Phys 224 Problem Set 3

Due October 8 at 11:59 PM. You may talk/consult with whom or whatever, but you must write your own solutions for submission.

1) You attend a local band's concert and hear a chord containing notes at 174.61 Hz and 508.35 Hz. How many half-steps above/below A4 are these notes? What band are you listening too?

2) The same band plays a chord with a C#2 and a note a quarter step above it. How many beats per second do you hear?

3) The vibrating length of strings on a viola is about 36 cm. What is the highest frequency fundamental a viola with steel strings could have? (reminder, the ultimate tensile strength of piano wire is 2.8 billion pascals, and the density is 7850 kg/m^3 .) What concert pitch (*i.e.* A4, B5 *etc.*) is this?

4) We saw in a demo that a guitar string deflects from its rest position by about 3mm horizontally (in each direction) when played. If the vibrating length of a guitar string is 63cm, what is the maximum change in angle the string makes with the bridge? If the tension in the string T is about 100N, what is force parallel to the bridge? It is this force that changes, and drives the production of sound. You may assume the string forms a triangle with height 3mm in the centre. See Figure 1 for a diagram.

5) You're walking by a construction site and see a straight hollow pole has been sunk into the ground. You can't see the bottom of the pole, and you don't know where the dirt level inside it is, but for some reason you develop a burning desire to know. You decide to blow across the top of the pole, and it produces a note at 34 Hz. How long is the hollow part of the hole? Speed of sound in air during nice weather is about 340 m/s .

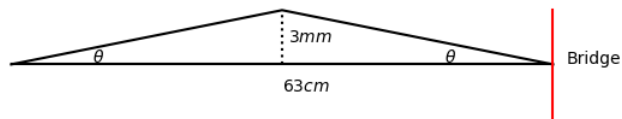


Figure 1: Bent guitar string for problem 4. The guitar string deviates in the centre by 3mm at maximum. The bridge is to the right. The left-right force along the string remains nearly constant, but the force parallel to the bridge (red vertical line) changes.